



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 24.09.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 23.09.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 23-सितंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 23.09.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 24-Sep-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	72711	63003	48358	28325	3445	215842
Peak Shortage (MW)	2493	0	0	544	0	3037
Energy Met (MU)	1703	1497	1192	643	74	5109
Hydro Gen (MU)	323	123	155	88	27	717
Wind Gen (MU)	10	62	176	-	-	248
Solar Gen (MU)*	167.31	87.58	114.59	2.92	1.15	374
Energy Shortage (MU)	17.48	1.08	0.00	2.59	0.00	21.15
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	77092	66880	57182	30204	3548	230608
Time Of Maximum Demand Met	12:24	14:48	11:34	00:01	19:35	14:28

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.055	0.38	1.33	13.01	14.72	73.00	12.28

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	15312	0	298.1	174.5	0.9	716	0.00
	Haryana	11850	0	245.8	177.0	-1.1	182	0.00
	Rajasthan	15722	0	329.2	112.3	2.4	815	13.32
	Delhi	6360	0	131.3	118.4	-0.3	305	0.00
	UP	26443	85	543.4	225.5	-1.4	238	1.05
	Uttarakhand	2393	0	50.4	25.0	0.1	145	1.22
	HP	1813	0	38.6	8.0	-0.4	107	0.00
	J&K(UT) & Ladakh(UT)	2587	178	54.3	29.9	5.8	504	1.89
	Chandigarh	370	0	7.4	6.5	0.9	91	0.00
WR	Railways_NR ISTS	200	0	4.1	3.7	0.5	67	0.00
	Chhattisgarh	5573	0	126.1	76.3	-0.8	390	0.00
	Gujarat	24205	0	494.4	185.0	-2.5	795	0.00
	MP	12596	0	271.5	136.4	-0.3	831	0.12
	Maharashtra	24387	861	536.0	172.9	0.8	950	0.96
	Goa	675	0	14.5	11.6	2.5	67	0.00
	DNHDDPDCL	1355	0	30.7	30.6	0.1	63	0.00
	AMNSIL	513	0	10.9	1.9	0.5	174	0.00
	BALCO	527	0	12.6	12.6	0.0	3	0.00
SR	Andhra Pradesh	10798	0	223.9	68.5	0.0	951	0.00
	Telangana	12202	0	243.2	93.5	-0.5	605	0.00
	Karnataka	14099	0	248.5	86.2	-2.3	652	0.00
	Kerala	4174	0	84.5	49.3	1.0	367	0.00
	Tamil Nadu	18095	0	381.4	131.8	-3.3	737	0.00
	Puducherry	492	0	10.6	10.3	-0.3	22	0.00
ER	Bihar	8001	10	165.6	157.2	-3.4	471	2.59
	DVC	3400	0	72.7	-52.1	-3.2	645	0.00
	Jharkhand	1965	0	44.5	34.0	0.3	172	0.00
	Odisha	5711	0	116.0	21.7	-1.9	351	0.00
	West Bengal	11576	0	243.0	102.2	-2.7	689	0.00
	Sikkim	96	0	1.5	1.4	0.1	20	0.00
	Railways_ER ISTS	22	0	0.2	0.1	0.0	4	0.00
NER	Arunachal Pradesh	192	0	4.4	3.1	0.9	52	0.00
	Assam	2355	0	52.2	44.6	0.5	196	0.00
	Manipur	190	0	1.1	2.8	-1.7	30	0.00
	Meghalaya	254	0	4.3	1.2	-0.2	144	0.00
	Mizoram	129	0	2.2	0.7	-0.1	16	0.00
	Nagaland	173	0	3.2	2.8	0.0	20	0.00
	Tripura	375	0	6.6	6.1	0.4	94	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	35.4	15.4	-12.2	-24.9
Day Peak (MW)	1787.0	820.9	-536.0	-1124.9

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	374.4	-256.4	-41.3	-89.4	12.6	0.0
Actual(MU)	363.4	-239.2	-43.8	-96.7	12.3	-3.9
O/D/U/D(MU)	-11.1	17.2	-2.5	-7.3	-0.3	-3.9

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3815	15165	4258	5216	259	28713	58
State Sector	4776	7759	6872	1534	155	21095	42
Total	8591	22924	11130	6750	414	49809	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	844	1472	699	704	18	3737	68
Lignite	28	17	57	0	0	102	2
Hydro	323	123	155	88	27	717	13
Nuclear	29	53	73	0	0	155	3
Gas, Naptha & Diesel	25	53	5	0	23	107	2
RES (Wind, Solar, Biomass & Others)	183	152	318	5	1	659	12
Total	1433	1870	1308	797	68	5477	100
Share of RES in total generation (%)	12.77	8.13	24.30	0.67	1.69	12.04	
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	37.35	17.56	41.80	11.70	40.95	27.97	

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.018
Based on State Max Demands	1.071

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	230608	14:28	30
Non-Solar hr	218677	19:00	5134

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 24-Sep-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	351	0.0	8.6	-8.6
2	HVDC	PUSAULI B/B	-	0	112	0.0	2.6	-2.6
3	765 kV	GAYA-VARANASI	2	221	534	0.0	2.6	-2.6
4	765 kV	SASARAM-FATEHPUR	1	27	249	0.0	2.2	-2.2
5	765 kV	GAYA-BALIA	1	0	873	0.0	15.8	-15.8
6	400 kV	PUSAULI-VARANASI	1	0	127	0.0	2.3	-2.3
7	400 kV	PUSAULI -ALLAHABAD	1	16	58	0.0	0.4	-0.4
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	575	0.0	8.9	-8.9
9	400 kV	PATNA-BALIA	2	0	885	0.0	17.5	-17.5
10	400 kV	NAUBATPUR-BALIA	2	0	476	0.0	5.1	-5.1
11	400 kV	BIHARSHARIFF-BALIA	2	2	176	0.0	2.7	-2.7
12	400 kV	MOTIHARI-GORAKHPUR	2	0	379	0.0	6.8	-6.8
13	400 kV	BIHARSHARIFF-VARANASI	2	218	276	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	140	0.0	0.5	-0.5
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.6	76.0	-75.4
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	0	1694	0.0	17.1	-17.1
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	2658	0	39.7	0.0	39.7
3	765 kV	JHARSUGUDA-DURG	2	0	717	0.0	9.1	-9.1
4	400 kV	JHARSUGUDA-RAIGARH	4	0	676	0.0	10.5	-10.5
5	400 kV	RANCHI-SIPAT	2	542	0	6.5	0.0	6.5
6	220 kV	BUDHIPADAR-RAIGARH	1	0	152	0.0	2.7	-2.7
7	220 kV	BUDHIPADAR-KORBA	2	0	206	0.0	3.2	-3.2
ER-WR						46.2	42.6	3.6
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	206	0.0	4.9	-4.9
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	2	0.0	0.0	0.0
3	765 kV	ANGUL-SRIKAKULAM	2	0	1723	0.0	26.6	-26.6
4	400 kV	TALCHER-I/C	2	1404	0	32.6	0.0	32.6
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	31.5	-31.5
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	0	469	0.0	7.7	-7.7
2	400 kV	ALIPURDUAR-BONGAIGAON	2	0	682	0.0	12.1	-12.1
3	220 kV	ALIPURDUAR-SALAKATI	2	0	152	0.0	2.7	-2.7
ER-NER						0.0	22.4	-22.4
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	502	0.0	12.0	-12.0
NER-NR						0.0	12.0	-12.0
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	4035	0.0	15.2	-15.2
2	HVDC	VINDHYACHAL B/B	-	0	499	0.0	10.3	-10.3
3	HVDC	MUNDRA-MOHINDERGARH	2	0	2353	0.0	55.9	-55.9
4	765 kV	GWALIOR-AGRA	2	0	1841	0.2	32.8	-32.6
5	765 kV	GWALIOR-PHAGI	2	0	2076	0.0	31.8	-31.8
6	765 kV	JABALPUR-ORAI	2	0	1385	0.0	49.5	-49.5
7	765 kV	GWALIOR-ORAI	1	727	0	10.8	0.0	10.8
8	765 kV	SATNA-ORAI	1	0	996	0.0	20.2	-20.2
9	765 kV	BANASKANTHA-CHITORGARH	2	1857	58	20.8	0.0	20.8
10	765 kV	VINDHYACHAL-VARANASI	2	0	4089	0.0	74.3	-74.3
11	400 kV	ZERDA-KANKROLI	1	382	71	3.9	0.1	3.7
12	400 kV	ZERDA -BHINMAL	1	426	212	0.1	2.5	-2.5
13	400 kV	VINDHYACHAL -RIHAND	1	0	1	0.0	0.0	0.0
14	400 kV	RAPP-SHUJALPUR	2	94	628	0.1	7.0	-6.9
15	400 kV	NEEMUCH-Chittorgarh	2	0	714	0.0	9.8	-9.8
16	220 kV	BHANPURA-RANPUR	1	0	88	0.0	1.5	-1.5
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.5	-2.5
18	220 kV	MEHGAON-AURAIYA	1	98	30	0.7	0.0	0.6
19	220 kV	MALANPUR-AURAIYA	1	66	48	0.3	0.3	0.0
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						36.8	313.6	-276.8
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	810	0.0	11.9	-11.9
2	HVDC	RAIGARH-PUGALUR	2	0	606	0.0	14.6	-14.6
3	765 kV	SOLAPUR-RAICHUR	2	1920	0	28.1	0.0	28.1
4	765 kV	WARDHA-NIZAMABAD	2	763	928	5.2	4.8	0.4
5	765 kV	WARORA-WARANGAL(NEW)	2	751	1010	4.2	5.7	-1.5
6	400 kV	KOLHAPUR-KUDGI	2	1923	0	37.5	0.0	37.5
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	85	0	1.5	0.0	1.5
9	220 kV	XELDEM-AMBEWADI	1	0	107	2.2	0.0	2.2
WR-SR						78.7	37.0	41.8

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	633	508	565	13.55	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	844	563	728	17.46	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	134	60	100	2.39	
	NER	132kV GELEPHU-SALAKATI	31	0	25	0.60	
	NER	132kV MOTANGA-RANGIA	70	0	60	1.44	
NEPAL	NR	NEPAL IMPORT (FROM UP)	-49	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	64	0	32	0.76	
	ER	NEPAL IMPORT (FROM BIHAR)	107	0	79	1.90	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	650	268	532	12.76	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-449	-380	-437	-10.49	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1125	-1001	-1039	-24.93	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-87	0	-73	-1.74	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10
Bangladesh	10.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.82
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	10.92	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.92

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	33.41	0.00	1.49	0.00	0.00	0.04	0.00	0.00	34.94
Nepal	7.35	0.00	8.74	0.00	0.00	0.00	0.00	0.00	16.09
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	40.76	0.00	10.23	0.00	0.00	0.04	0.00	0.00	51.03

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	33.41	0.00	1.49	0.00	0.00	0.04	0.00	0.00	34.94
Nepal	7.25	0.00	8.74	0.00	0.00	0.00	0.00	0.00	15.99
Bangladesh	-10.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-10.82
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	29.84	0.00	10.23	0.00	0.00	0.04	0.00	0.00	40.11